Pre-work · Nervous Tissue and Neuronal Integration

Functional Organization and Nervous Tissue, and Gross Anatomy and Neuronal Integration. Read both Part A chapters first. This packet does not repeat them. Every other packet in Module 5 is built on the cell and the vocabulary in these two chapters, so the time you spend here is repaid four more times.

Module 5 · 1 of 5 | Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1. **Start with the reading.**
   Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.
2. **Organize what you read.**
   In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.
3. **Draw it.**
   In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.
4. **Apply it.**
   Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.
5. **Come back to it later in the week**
   , not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

3 min

PANEL 1 OF 3

Turn the two tissue chapters into mini visuals. Eight chunks. The grids carry the naming pairs, the hubs carry the structure, and the one chain is the reflex arc.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 **HUB**

The nervous system and its divisions

Nervous system at the hub. Spoke first to the CNS with the brain and spinal cord, then to the PNS with nerves, ganglia, and sensory receptors. Off the motor side of the PNS hang somatic, autonomic, and enteric, and off autonomic hang sympathetic and parasympathetic. Write the effector each motor division reaches at the end of its spoke.

2 **HUB**

The neuron

One cell body in the middle. Spoke out to dendrites, axon hillock, axon with axoplasm and axolemma, and axon terminals. Inside the cell body mark the nucleus, the Nissl bodies, the neurofibrils, and the microtubules. Draw an arrow along the whole cell showing which way an impulse travels, because the direction is what names the parts.

3 **GRID**

Classifying neurons, twice

Neuron type down the side: multipolar, bipolar, unipolar. Number of processes, where it is found, and which functional class usually wears that shape across the top. Then add two rows at the bottom for Purkinje and pyramidal cells and write where each one lives.

4 **GRID**

The six neuroglia

Glial cell down the side, all six. CNS or PNS, and the job it does across the top. Add a third column: what goes wrong if this cell fails. Four rows are CNS and two are PNS, and keeping them grouped is half the learning.

5 **SPLIT**

PNS myelination against CNS myelination

One split. Which cell builds the sheath, how many axons one cell can reach, neurolemma present or absent, and whether a cut axon can regrow. This is the pair that answers the regeneration question, so make the neurolemma row impossible to miss.

6 **SPLIT**

Gray matter against white matter

One split. What each is made of, and where each sits. Then hang two small sketches off the branches: a spinal cord cross-section with gray inside, and a brain section with gray outside. The swap between the two organs is the whole point of the visual.

7 **GRID**

Tract, nerve, nucleus, ganglion

A true two by two. Bundle of axons and cluster of cell bodies down the side, CNS and PNS across the top. Write the term in each of the four cells, then add one example of each. If you can fill this grid blank you have four terms for the price of one.

8 **CHAIN**

The reflex arc, five components

Five boxes with arrows, receptor through to effector. Write beside each box where that component physically sits, and mark the one box that is inside the CNS. Then note beside the integration center whether the arc is monosynaptic or polysynaptic.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

3 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

DRAWING 1

A multipolar neuron, fully labelled

BRAIN DUMP FIRST

Two minutes. Every part of the cell, and the direction a signal moves through it.

Draw one large multipolar neuron. Give it many branched dendrites, one cell body with the nucleus and Nissl bodies inside, a clear axon hillock, and one long axon ending in branched terminals with synaptic end bulbs. Wrap the axon in a myelin sheath with nodes of Ranvier between the segments. Now enlarge one end bulb in a bubble and draw the synaptic vesicles, the cleft, and the next cell's membrane.

In your notebook, head the page **M5-4 A multipolar neuron, fully labelled** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can put your finger on any part of your drawing and say whether a signal at that point is travelling toward the cell body or away from it.

DRAWING 2

A myelinated axon in the PNS and in the CNS, side by side

BRAIN DUMP FIRST

One minute. Which cell wraps each one, and what is left behind on the outside.

Draw two axons beside each other. On the left, wrap the PNS axon with Schwann cells, one per segment, and draw the neurolemma as the outer layer of each Schwann cell. On the right, draw one oligodendrocyte with several arms, each arm wrapping a segment of a different CNS axon, and leave the outer surface bare. Mark the nodes of Ranvier on both and draw the impulse jumping node to node.

In your notebook, head the page **M5-5 A myelinated axon in the PNS and in the CNS, side by side** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can explain from your two drawings why a cut nerve in the forearm can regrow and a cut tract in the cord usually cannot.

DRAWING 3

Two reflexes drawn as arcs

BRAIN DUMP FIRST

Two minutes. All five components in each, and the difference in the middle.

Draw a spinal cord cross-section twice. On the first, draw the patellar reflex: the muscle spindle receptor, the sensory neuron entering the posterior root, one synapse in the cord, the motor neuron leaving the anterior root, and the quadriceps as the effector. On the second, draw the withdrawal reflex with an interneuron added between the sensory and motor neurons. Label all five components on both and mark the posterior root ganglion.

In your notebook, head the page **M5-6 Two reflexes drawn as arcs** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can count the synapses on each of your drawings and name the reflex type from the count alone.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

A patient cuts a nerve in the forearm and, over many months, sensation and movement return. Another patient has a tract severed in the spinal cord and recovers nothing. Both injuries cut axons. Explain the difference from structure.

Answer. The forearm nerve is myelinated by Schwann cells, which leave a neurolemma. The cord tract is myelinated by oligodendrocytes, which do not.

Reasoning. Start with the cell that builds the sheath, not with the axon. In the PNS a Schwann cell wraps one segment of one axon, and its outer layer, the neurolemma, survives the injury as a hollow tube. That tube is a guide, so the regrowing axon has a path back to its target. In the CNS one oligodendrocyte reaches up to fifteen axons and leaves no neurolemma behind, so a regrowing axon has nothing to follow. The location of the injury also matters, because an axon cut off from its cell body loses its supply of proteins and dies beyond the cut. So the answer is not about the axon at all, it is about which glial cell was doing the wrapping.

Set A · Name it

1. The gaps in the myelin sheath along an axon are the ________________________.
2. A cluster of neuron cell bodies located outside the CNS is a ________________________.
3. The glial cell that lines the ventricles and helps circulate cerebrospinal fluid is the ________________________.
4. The cone-shaped region where the cell body joins the axon is the ________________________.
5. Name the five components of a reflex arc, in order.

Set B · Predict it

1. The immune system strips myelin from axons in the CNS. Predict the effect on impulse speed and name the cell that was lost.
2. An axon is cut halfway along its length. Predict what happens to the portion beyond the cut and explain why, using the cell body.
3. A neuronal pool is wired for divergence. Predict what one incoming signal does to the number of neurons responding.
4. A stroke damages a descending pathway on the right side of the brain. Predict which side of the body is weakened and name the reason.

Set C · Tell it apart

1. Distinguish a tract from a nerve, and a nucleus from a ganglion, by what each is made of and where it sits.
2. Explain why a tumor of nervous tissue almost always arises from glia rather than from neurons.
3. Distinguish a monosynaptic from a polysynaptic reflex by the wiring in the integration center, and give an example of each.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

Mini visual sheet

2 min

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

M5-1

The nervous system and its divisions

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

M5-2

The neuron

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

M5-3

Classifying neurons, twice

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

M5-4

The six neuroglia

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

M5–5

PNS myelination against CNS myelination

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

M5–6

Gray matter against white matter

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

M5-7

Tract, nerve, nucleus, ganglion

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

M5-8

The reflex arc, five components

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.